ETALE-LOCAL UNIQUENESS OF LIFTS THROUGH A THICKENING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1210-1232 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1234-1303, the complete author proof (68 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_stacks-stacks-more-morphisms.json (pb=2807, stmt 1210-1232, proof 1234-1303).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 09ZU: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 364–373. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CF4: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 711–747. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1303–1303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CHJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1303–1303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{X} \subset \mathcal{X}'$ be a thickening of algebraic stacks. Consider a commutative diagram*

$$\begin{array}{ccccc} W'' & \longleftarrow & W & \longrightarrow & W' \\ \downarrow{\scriptstyle x''} & & \downarrow{\scriptstyle x} & & \downarrow{\scriptstyle x'} \\ \mathcal{X}' & \longleftarrow & \mathcal{X} & \longrightarrow & \mathcal{X}' \end{array}$$

with cartesian squares where W', W, W'' are algebraic spaces and the vertical arrows are smooth. Then there exist

(1) *an étale covering $\{f'_k : W'_k \to W'\}_{k \in K}$,*
(2) *étale morphisms $f''_k : W'_k \to W''$, and*
(3) *2-morphisms $\gamma_k : x'' \circ f''_k \to x' \circ f'_k$*

such that (a) $(f'_k)^{-1}(W) = (f''_k)^{-1}(W)$, (b) $f'_k|_{(f'_k)^{-1}(W)} = f''_k|_{(f''_k)^{-1}(W)}$, and (c) pulling back γ_k to the closed subscheme of (a) agrees with the 2-morphism given by the commutativity of the initial diagram over W.

Proof. Denote $i : W \to W'$ and $i'' : W \to W''$ the given thickenings. The commutativity of the diagram in the statement of the lemma means there is a 2-morphism $\delta : x' \circ i' \to x'' \circ i''$ This is the 2-morphism referred to in part (c) of the statement. Consider the algebraic space

$$I' = W' \times_{x', \mathcal{X}', x''} W''$$

with projections $p' : I' \to W'$ and $q' : I' \to W''$. Observe that there is a "universal" 2-morphism $\gamma : x' \circ p' \to x'' \circ q'$ (we will use this later). The choice of δ defines a morphism

$$\begin{array}{ccccc} W & & \xrightarrow{\quad\delta\quad} & & I' \\ & \searrow^{i'} & & \swarrow_{i''} & \\ & & W' \times_{x', \mathcal{X}', x''} W'' & & \end{array}$$

such that the compositions $W \to I' \to W'$ and $W \to I' \to W''$ are $i : W \to W'$ and $i' : W \to W''$. Since x'' is smooth, the morphism $p' : I' \to W'$ is smooth as a base change of x''.

Suppose we can find an étale covering $\{f'_k : W'_k \to W'\}$ and morphisms $\delta_k : W'_k \to I'$ such that the restriction of δ_k to $W_k = (f'_k)^{-1}$ is equal to $\delta \circ f_k$ where $f_k = f'_k|_{W_k}$. Picture

$$\begin{array}{ccccc} W_k & \xrightarrow{f_k} & W & \xrightarrow{\delta} & I' \\ \downarrow & \nearrow_{\delta_k} & & & \downarrow{\scriptstyle p'} \\ W'_k & & \xrightarrow{f'_k} & & W' \end{array}$$

In other words, we want to be able to extend the given section $\delta : W \to I'$ of p' to a section over W' after possibly replacing W' by an étale covering.

If this is true, then we can set $f''_k = q' \circ \delta_k$ and $\gamma_k = \gamma \star \mathrm{id}_{\delta_k}$ (more succinctly $\gamma_k = \delta_k^* \gamma$). Namely, the only thing left to show at this is that the morphism f''_k is étale. By construction the morphism $x' \circ p'$ is 2-isomorphic to $x'' \circ q'$. Hence $x'' \circ f''_k$ is 2-isomorphic to $x' \circ f'_k$. We conclude that the composition

$$W'_k \xrightarrow{f''_k} W'' \xrightarrow{x''} \mathcal{X}'$$

is smooth because $x' \circ f'_k$ is so. As f_k is étale we conclude f''_k is étale by Lemma 0CF4.

If the thickening is a first order thickening, then we can choose any étale covering $\{W'_k \to W'\}$ with W'_k affine. Namely, since p' is smooth we see that p' is formally

smooth by the infinitesimal lifting criterion (More on Morphisms of Spaces, Lemma 04AM). As W_k is affine and as $W_k \to W'_k$ is a first order thickening (as a base change of $\mathcal{X} \to \mathcal{X}'$, see Lemma 09ZU) we get δ_k as desired.

In the general case the existence of the covering and the morphisms δ_k follows from More on Morphisms of Spaces, Lemma 0CHJ. □